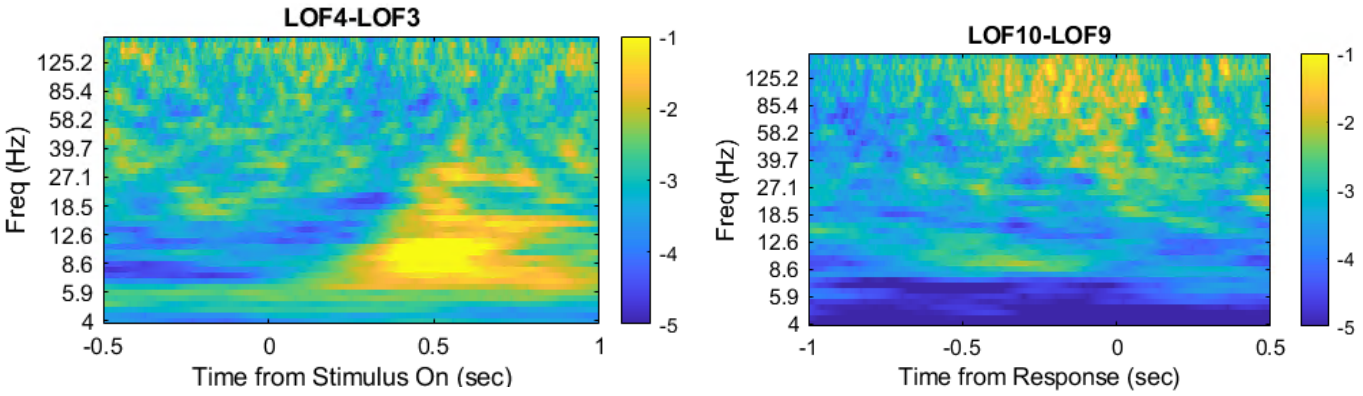
Spring 2026

BMEN 5413 Homework #4-LFP Processing

Download the Matlab data file (HW4Data.mat) from Canvas. In this file you will find several variables:

- **bipolarContactInfo**: contact info as table for each recorded bipolar channel.
- **thetaHaa & highGammaHaa**: Hilbert Analytic Amplitude (HAA) of Theta band (4-7.5 Hz) activity or High Gamma band (62-170 Hz) activity aligned to stimulus onset (top row) or response time (bottom row). Each column is a different channel.
    - Within each element of the cell array is a matrix of data. In the matrix each row corresponds to a different trial and columns correspond to time.
- **erpParametersStimulus & erpParametersResponse**: the stimulus or response event-alignment parameters.
    - The field timeWindow indicates the relative time in *ms* to the event corresponding to the columns inside each cell of the HAA data.
- **eventTableStimulus & eventTablesResponse**: event times, event types, and other information corresponding to each trial in the HAA data.
    - conflictLevel: conflict level as low (0) or high (1)
    - reactionTime: reaction time in seconds (NaNs -> no response)
    - correctResponse: whether the participant was correct (1) or incorrect (0)
    - Stimulus: the numerical stimulus shown to participants.
- **fs**: sampling rate (500 Hz).



The goal of this homework assignment is to do some relatively simple LFP analysis using the same data that you used in homework 3. A visual inspection of the spectrograms from homework 3 (see figures above) aligned to stimulus onset or response time reveals evidence of Theta band and High Gamma band activity on average. While it is possible to run statistical analysis on the spectrogram data directly, there are two major issues that arise from this. First, High Gamma is often broadband with individual bursts of High Gamma activity occupying slightly different frequencies at slightly different times. Second, analysis of spectrograms is computationally expensive (e.g. 1000 ms x 50 Hz-bins). To address these issues, I provided you with the Hilbert Analytic Amplitude (HAA) for Theta (4-7.5 Hz) and High Gamma (62-170 Hz) bands extracted from the bipolar re-referenced data.

Data and code for this homework assignment comes from the Herman-Darrow lab. Sharing of this data outside of the homework without prior permission is not allowed!

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For this homework assignment, you will be analyzing the channel-level encoding effect sizes using ROC analysis and group-level encoding using Linear Models. Specifically, we will be analyzing the effects of conflict level and reaction time on HAA data. For the first part we will ignore reaction time since reaction time is a continuous variable.

Part 1: Channel-Level Encoding Effect Sizes:

- For this part you must analyze data for *each* bipolar channel separately.
- Plot the average stimulus-aligned and response-aligned HAA activity for each channel.
    - It would be helpful to create 4 subplots, 1 for each event and band and plot the activity of all 9 bipolar channels on a single subplot.
    - The units of the HAA activity are arbitrary or z-scored.
- What do you see in the average HAA activity? Do responses appear better aligned to stimulus onset or response time? Or both?
- Pick a single bipolar channel (e.g. channel 9) and plot the average response of that channel either aligned to stimulus onset or response time for Theta and High Gamma bands. Then pick a single time point (e.g. 200 ms before response time) and make a histogram of the trial-by-trial HAA values. Do the histograms look normally distributed?
- Calculate the Area Under the Curve (AUC) for each bipolar channel, at each time point, and for each band assuming that you can discriminate between conflict levels using the HAA activity. This can be calculated easily using perfcurve in Matlab.
- Plot the AUC curves over time for the data above. Again, using 4 subplots would be helpful. Note you may get values less than 0.5 which indicate the encoding is in the opposite direction.
- What do you observe in the data? Could you get good classification with individual bipolar channels? What is the best discrimination that you got (AUC farthest from 0.5)? Which band was it? Was it aligned to response time or stimulus onset?
- Calculate d' (d-prime) from the AUC values and plot these. D-prime is a common effect size measure that can be calculated using means and standard deviations or from AUC values (dPrime = sqrt(2)*norminv(AUC)). Note, you may want to convert your AUC values < 0.5 to values greater than 0.5 before calculating d-prime!
- While there are different ways of interpreting d-prime values, 0.2 can be used as a threshold for small effect sizes, 0.5 for medium effect sizes, and 0.8 for large effect sizes. How would you interpret the effect size of this data?

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Linear [regression] Model In Wilkinson's Notation:
HAA ~ conflictLevel + reactionTime

Part 2: Group-level encoding using Linear Models:

- In this part we are going to *group all of the channels together* into a single group-level model. Normally, we might do this across many channels in a single brain region (e.g. DLPFC) across multiple participants, but this would require a lot of data and computational power.
    - Keep the stimulus onset and response aligned data separate.
    - Keep theTheta and High Gamma band activity separate.
- We will be using Wilkinson's notation in Matlab to build Linear Models (see notation above).
- For each event and band separately, group all the data together into an event table and an event data matrix. We need to concatenate/stack the data on top of each other, but in an organized manner.
- Fit each time point in each group with a Linear Model (fitlm.m).
    - Save the t-statistic for each fixed effect (i.e. conflictLevel and reactionTime) and p-value.
- Plot the time course of the t-statistics for each fixed effect in each group-level model. What do you notice? Since the number of data points is the same across models, we can make some inferences about the results. Do these results differ from the channel-level results?
- Choose a single band and event and run a statistical test on the p-values from the linear models using False Discovery Rate (FDR) correction (fdr_bh.m). Note, we would normally run statistics on all events and bands instead of picking one. Also, it may be easier to calculate the new FDR adjusted p-value threshold (crit_p instead of 0.05) than the FDR adjusted p-values (adj_p). Plot your statistically significant results on top of the t-statistic curves to visualize the outcome.
- High Gamma signals look noisier than the Theta signals. Name one way you rectify this?